

comes in. Failing to comply with data privacy laws (like GDPR or CCPA), intellectual property rights, or other relevant regulations can lead to hefty fines and legal trouble, blowing your budget out of the water. So, compliance isn't just a legal necessity; it's a financial one.

Factor compliance costs into your budget from the start. This might include legal consultations, data anonymisation tools, or security audits. By considering both budget and compliance together, you can ensure your LLM project is not only innovative but also financially sound and legally secure.

TABLE 2 A COMPARISON OF PYTORCH AND TENSORFLOW		
Feature	PyTorch	TensorFlow
Flexibility	Dynamic computation graph for flexibility	Optimised for large-scale production deployments
Adoption	Popular among researchers and academics	Backed by Google with strong enterprise adoption
Ecosystem	Strong Hugging Face integration	Extensive tools for model optimisation and serving

TABLE 3 OTHER OPEN SOURCE LLM FRAMEWORKS	
EleutherAI (GPT-Neo, GPT-J, GPT-NeoX)	Open source alternatives to OpenAI's GPT models.
Cohere	Provides API-based access to large language models.
BLOOM	A multilingual open source LLM developed by the BigScience initiative.

TABLE 4 TOOLS FOR PREPROCESSING AND MANAGING TRAINING DATA		
Category	Description	Popular tools
Data cleaning and annotation	Before training a model, data must be cleaned and properly labelled. This ensures accuracy and reduces biases in machine learning models. Annotation tools help in tagging datasets for supervised learning.	Prodigy (AI-assisted annotation), Label Studio (open source data labelling), Snorkel (programmatic data labelling)
Dataset management	Keeping track of dataset versions, changes, and metadata is crucial for reproducibility in ML projects. Dataset management tools help maintain consistency and collaboration.	Weights & Biases (W&B) (experiment tracking and dataset versioning), DVC (Data Version Control, Git-like versioning for datasets)
Synthetic data generation	When real-world data is insufficient or lacks diversity, synthetic data generation can create realistic, model-ready datasets. AI-powered tools can generate high-quality training data for various applications.	GPT-based generators (AI-generated synthetic data), Other tools (custom solutions for domain-specific needs)

POPULAR OPEN SOURCE FRAMEWORKS FOR BUILDING LLMs

Hugging Face Transformers

Hugging Face Transformers is one of the most popular open source libraries for working with LLMs. It provides an easy-to-use interface for deploying and fine-tuning transformer models, making it an essential tool for researchers and developers.

Strengths.

- Vast model repository with thousands of pretrained models.
- Simple APIs for inference and training.
- Strong community support and extensive documentation.
- Supports multiple frameworks, including PyTorch and TensorFlow.

Use cases.

- Text generation and summarisation
- Sentiment analysis
- Question answering and conversational AI

Hugging Face hosts a vast collection of pretrained models, such as BERT, GPT-2, T5, and BLOOM. Users can fine-tune these models on custom datasets using transfer learning, reducing computational

costs compared to training from scratch.

OpenAI's GPT models (via open source APIs)

Although OpenAI's latest GPT models are not fully open source, it provides APIs for accessing models like GPT-4 and GPT-3.5. Community-driven implementations, such as GPT-J by EleutherAI, attempt to provide open alternatives.

Key points.

- OpenAI's API provides access to powerful models but requires usage-based licensing.
- Open implementations like GPT-NeoX offer community-driven alternatives.
- GPT models excel in text completion, summarisation, and conversational AI applications.

PyTorch and TensorFlow

PyTorch and TensorFlow are foundational deep-learning frameworks used for developing LLMs from scratch or fine-tuning existing ones. Table 2 compares them.

LangChain

LangChain is an open source framework designed to build LLM-powered applications by integrating different components like memory, tools, and agents.

Features.

- Simplifies building AI-powered workflows.
- Provides modules for context-aware reasoning and tool use.
- Compatible with multiple LLM